

KHOLM, Russian Federation

WMO: 263780

Lat: 57.150N

Lon: 31.183E

Elev: 233

StdP: 14.57

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-6.6	-18.2	2.1	-12.8	-11.3	3.0	-5.9	14.3	28.1	13.2	28.9	1.2	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	82.5	67.1	79.7	65.1	76.9	64.1	69.2	78.4	67.5	76.5	65.6	73.9	4.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	96.8	73.6	64.2	90.7	71.5	62.4	84.9	69.4	33.4	77.9	32.0	76.8	30.5	73.8	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.8	10.6	DB	-21.1	87.2	5.2	2.6	-24.8	89.0	-27.8	90.5	-30.7	92.0	-34.5	93.9	(4)
(5)			WB	-21.4	72.1	5.2	0.8	-25.1	72.7	-28.1	73.1	-31.0	73.6	-34.8	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	19.7	25.0	30.7	42.4	53.6	58.8	62.9	60.6	52.6	41.4	32.8	26.5		
	DBStd	16.96	12.71	11.45	9.37	7.59	7.71	6.71	5.57	5.78	6.92	7.96	8.06	10.75		
	HDD50	4122	940	701	597	250	48	11	0	2	50	279	516	728		
	HDD65	8361	1405	1121	1062	678	359	203	108	161	374	731	966	1193		
	CDD50	1329	0	0	0	23	160	275	399	331	128	13	0	0		
	CDD65	92	0	0	0	0	7	17	41	25	2	0	0	0		
	CDH74	1302	0	0	0	6	182	274	493	315	32	0	0	0		
	CDH80	216	0	0	0	0	35	38	91	51	1	0	0	0		
	WSAvg	5.1	5.2	5.4	6.1	5.2	5.0	4.7	4.1	4.2	4.5	5.3	5.7	5.8		
	PrecAvg	29.4	2.2	1.7	1.5	1.8	2.4	3.1	4.1	3.2	2.7	2.5	2.4	2.1		
(15) Precipitation	PrecMax	38.5	3.3	2.8	3.0	4.1	3.9	5.9	7.9	5.9	5.8	5.2	4.5	3.7		
	PrecMin	22.9	0.8	0.3	0.4	0.5	1.0	0.4	0.5	0.6	0.7	0.7	0.0	1.1		
	PrecStd	4.1	0.8	0.6	0.9	1.0	0.9	1.6	2.4	1.7	1.4	1.1	1.3	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	42.3	57.5	74.4	84.2	84.3	86.0	84.4	77.9	62.9	52.1	47.7		
		MCWB	40.3	37.7	46.6	53.9	65.1	65.2	68.8	67.6	65.1	55.4	51.0	45.3		
	2%	DB	38.8	39.3	50.4	68.2	79.0	80.0	82.5	81.2	73.4	57.8	48.5	43.1		
		MCWB	37.5	36.0	41.1	52.8	61.4	64.9	67.6	66.3	63.7	52.3	46.8	41.6		
	5%	DB	36.6	37.7	45.8	62.0	74.5	77.1	79.6	77.7	69.1	55.1	45.8	40.8		
		MCWB	34.9	35.1	39.0	49.5	59.2	63.2	66.3	65.4	60.6	51.5	44.1	39.4		
	10%	DB	34.7	36.3	42.3	57.7	70.0	74.0	76.2	73.9	65.2	53.2	43.1	37.8		
		MCWB	33.3	34.2	37.8	47.0	56.5	61.3	65.1	63.6	57.9	50.3	41.5	36.3		
	0.4%	WB	40.7	38.6	47.4	58.1	67.4	70.9	71.1	70.7	66.6	56.3	51.4	45.7		
		MCDB	41.6	41.2	56.1	68.7	80.4	78.8	80.2	79.9	76.0	60.4	51.6	47.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.7	36.8	42.7	53.8	64.0	68.0	69.5	68.4	63.8	53.7	47.3	41.8		
		MCDB	38.6	38.5	46.8	63.9	74.7	76.8	79.1	77.2	71.5	56.2	48.5	42.9		
	5%	WB	35.3	35.6	40.2	50.9	61.6	65.1	67.8	66.7	61.5	52.1	44.0	39.4		
		MCDB	36.5	37.3	44.3	61.3	70.4	72.5	76.8	74.9	68.0	54.7	45.5	40.8		
	10%	WB	33.5	34.6	38.2	47.8	59.5	63.0	65.7	64.6	59.1	50.6	41.7	36.5		
		MCDB	34.6	36.0	42.1	56.2	66.9	71.0	74.1	72.2	64.4	53.3	43.1	37.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.6	14.9	20.2	24.0	22.3	19.8	22.2	18.7	12.3	7.7	7.3		
		MCDBR	6.8	9.8	20.4	30.4	31.9	28.7	27.3	28.7	27.1	16.3	9.6	6.6		
		MCWBR	6.3	7.6	12.9	17.0	15.4	14.5	13.4	15.4	16.7	12.3	8.6	6.1		
	5% WB	MCDBR	6.6	7.0	16.7	26.6	25.8	23.6	24.2	24.9	24.3	14.7	9.2	6.1		
		MCWBR	6.2	5.9	11.5	15.5	13.0	12.8	12.7	14.2	15.3	11.7	8.6	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.321	0.343	0.389	0.380	0.374	0.398	0.403	0.366	0.345	0.325	0.283		
	taud		2.139	2.122	2.175	2.254	2.364	2.403	2.350	2.330	2.408	2.382	2.240	2.149		
	Ebn at Noon		179	223	253	260	272	275	264	254	247	218	164	146		
	Edh at Noon		17	27	34	37	35	34	36	34	27	22	16	13		
(44) All-Sky Solar Radiation	RadAvg		139	390	840	1234	1643	1723	1666	1328	841	373	126	79		
	RadStd		31	60	108	148	159	157	170	123	91	72	21	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-284	-326	N/A	N/A

Nomenclature: See separate page